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小苏打浓度与煮制时间对意大利面拉面化效果的影响：一项 3 × 3 探索性厨房实验

Effects of Baking Soda Concentration and Cooking Time on the Ramen-Style Transformation of Spaghetti: A 3 × 3 Exploratory Kitchen Trial

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Figure 1. Source photograph provided by the author.

摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房实验比较2、4、6 g/L小苏打与10、12、15分钟煮制时间的九种组合。4 g/L、15分钟得分最高（9.5/10）；4 g/L、12分钟得分8.5/10。低浓度变化有限，高浓度长时间处理出现苦涩、黏口和肥皂味。

Abstract: A 3 x 3 exploratory kitchen trial combined baking soda concentrations of 2, 4, and 6 g/L with cooking times of 10, 12, and 15 minutes. The highest score, 9.5/10, was assigned to 4 g/L for 15 minutes. Lower concentration produced limited transformation, while prolonged high-alkali cooking produced bitterness, stickiness, and a soap-like odor.

Keywords: baking soda; spaghetti; alkaline noodles; culinary methods

1. Background / 背景

Commercial dried spaghetti may serve as a practical substitute when fresh alkaline noodles are unavailable. This trial compared concentration-time combinations under controlled kitchen conditions.

2. Materials and Methods / 材料与方法

Each treatment used 1 L water over medium heat. Baking-soda concentrations were 2, 4, and 6 g/L and cooking times were 10, 12, and 15 minutes. A single evaluator scored elasticity, smoothness, alkaline aroma, and ramen-like similarity.



Figure 2. Commercial spaghetti and 4 g baking soda.

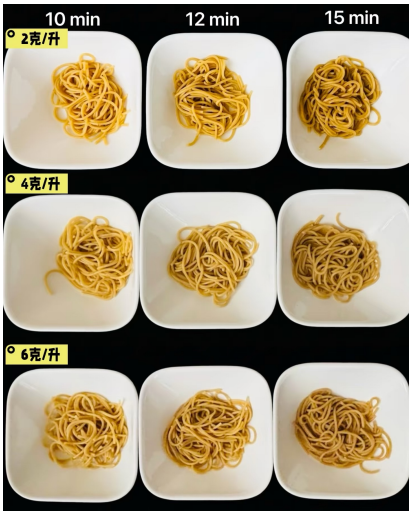


Figure 3. Nine concentration-time treatments.

3. Results / 结果

g/L	10 min	12 min	15 min
2	5.0	6.0	6.5
4	7.0	8.5	9.5
6	6.0	5.0	3.5



Figure 4. Original score matrix.

4. Practical Procedure / 实用方法

Add 4 g baking soda to 1 L water, bring to a boil, add spaghetti, and cook for 15 minutes. A 12-minute endpoint may suit a firmer preference. Use a larger pot and monitor foaming.

5. Discussion / 讨论

The pattern suggests a concentration-time interaction. More alkali did not necessarily produce a better ramen-like result.

6. Limitations / 局限性

This was a single-observer exploratory trial without independent replication, blinding, instrumental texture, pH, moisture, or inferential statistics.

7. Conclusion / 结论

Within this trial, 4 g/L for 15 minutes received the highest score.

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